No “true” greenery

Deciphering the bias of satellite and street view imagery in urban greenery measurement

Yingjing Huang[a,b,c]*, Rohit Priyadarshi Sanatani[c], Chang Liu[c], Yuhao Kang[d], Fan Zhang[b], Yu Liu[b], Fabio Duarte[c], Carlo Ratti[c]

[a]University of Vienna [b]Peking University [c]MIT Senseable City Lab [d]UT Austin

*yingjing.huang@univie.ac.at

Abstract

Urban greenery is a crucial element in building sustainable cities and communities. Despite the widespread use of satellite and street view imagery in monitoring urban greenery, there are significant discrepancies and biases in their measurement across different urban contexts. This study utilizes the Normalized Difference Vegetation Index (NDVI) from satellite imagery and the Green View Index (GVI) from street view imagery to measure urban greenery in ten global cities. By analyzing the distribution and visual differences of these indices, we identify **eight factors** causing measurement biases. A machine learning model is trained to estimate the bias risks across urban areas. We find that bias in most cities primarily stems from an **underestimation of GVI**, while Dubai and Seoul present fewer areas with overall bias risk. These findings emphasize the importance of carefully selecting and integrating these measurements for specific urban tasks, as there is no “true” greenery.

1 Introduction

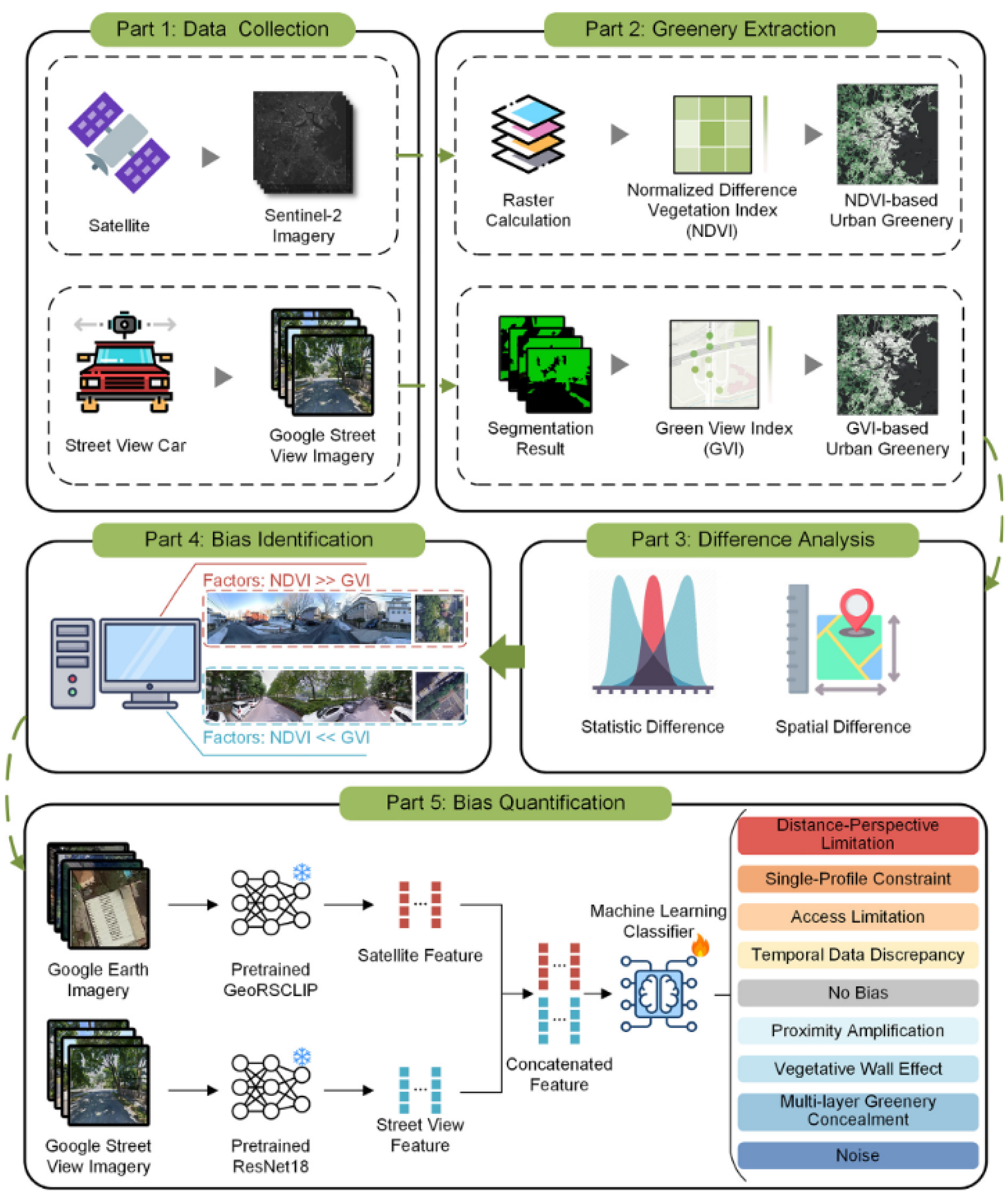
Urban greenery is fundamental to sustainable cities, impacting residents' well-being, air quality, biodiversity, and urban heat regulation. Two main approaches exist for large-scale measurement: **satellite imagery** (e.g., NDVI) offers broad top-down coverage but cannot capture vertical greenery; **street view imagery** (e.g., GVI) observes the vertical environment but is limited by obstructions and sparse sampling.

Both data sources have inherent limitations, yet these biases have **not been systematically discussed on a global scale**. This study addresses this gap by examining ten cities worldwide, identifying eight bias factors, and quantifying bias risks via machine learning.

2 Methods

Study areas: Ten cities – Amsterdam, Barcelona, Boston, Buenos Aires, Dubai, Johannesburg, Los Angeles, Melbourne, Seoul, and Singapore.

- **Data:** Sentinel-2 imagery (10-m) and 3.68M Google Street View images.
- **Greenery Extraction:** NDVI via raster calculation; GVI via DPT semantic segmentation. Both aggregated to 100-m grids.
- **Difference Analysis:** Getis–Ord Gi* hotspot analysis on normalized NDVI–GVI differences.
- **Bias Identification:** Visual annotation of 1,897 samples across ten cities to categorize eight bias factors.
- **Bias Quantification:** SVM model trained on GeoRSCLIP (satellite) and ResNet18 (street view) features to classify bias types across urban areas.



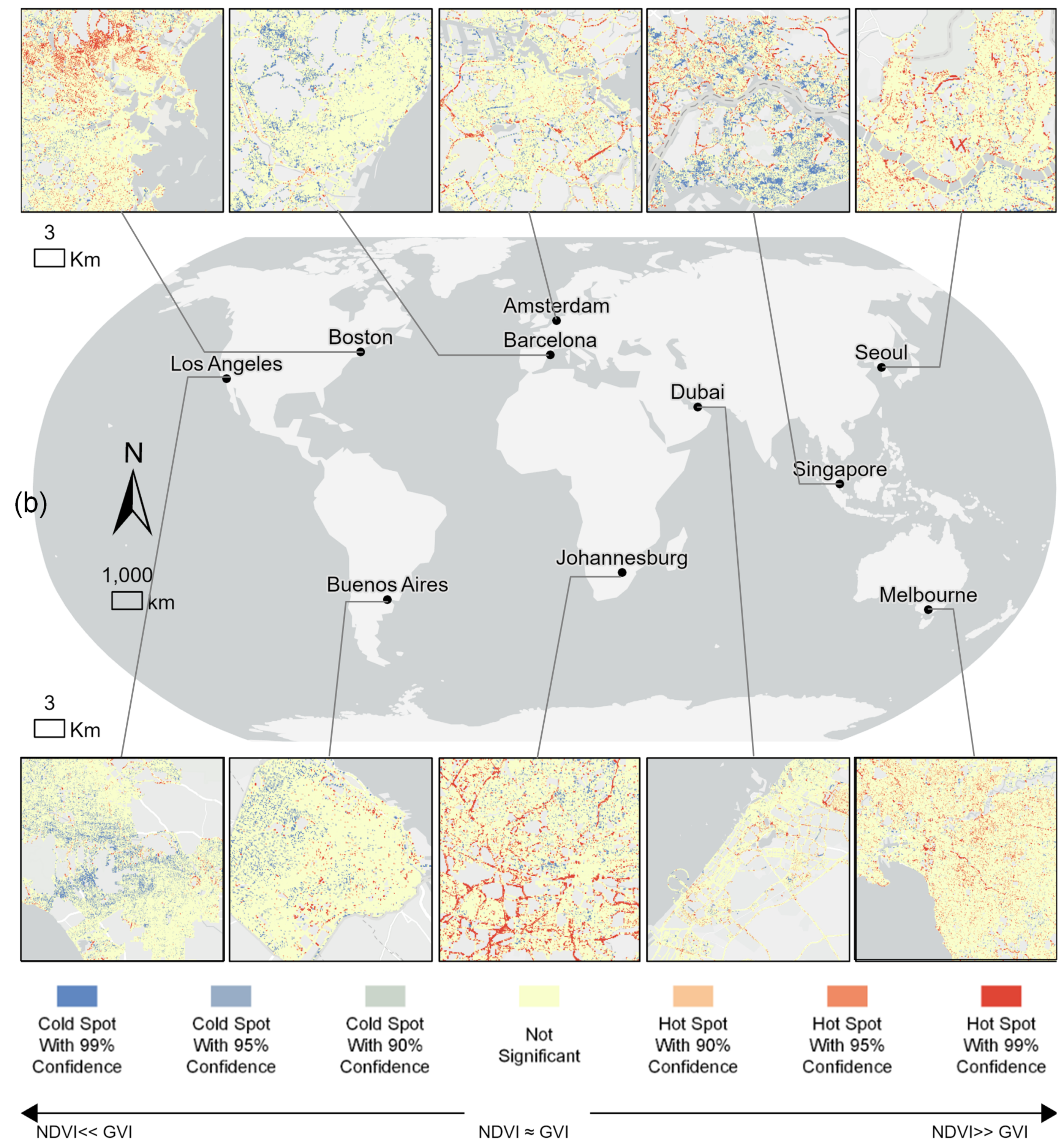
3 Results

NDVI and GVI show general alignment but notable differences across cities. Hotspot analysis reveals areas where GVI < NDVI display a **network-like distribution**, while GVI > NDVI areas concentrate in urban centers. We identified **eight primary bias factors**:

GVI underestimation:
1. Distance-perspective limitation
2. Single-profile constraint
3. Access limitation
4. Temporal data discrepancy

NDVI underestimation:
5. Proximity amplification
6. Vegetative wall effect
7. Multi-layer greenery concealment
8. Noise (cloud cover)

Bias in most cities stems from **GVI underestimation** (Barcelona being an exception). Dubai and Seoul show lowest bias risk; Amsterdam, Johannesburg, and Singapore show highest.



4 Conclusion

Neither satellite nor street view imagery alone captures the “true” state of urban greenery – integrating both is essential. The predominant bias type varies by city. Urban planners should account for these biases when using greenery data for decision-making.

References and Full Paper

[1] M. Helbich, ..., R. Schram, Can't see the wood for the trees? An assessment of street view- and satellitederived greenness measures in relation to mental health, Landsc. Urban Plan. 214 (2021) 104181.

[2] F. Biljecki, ..., Y. Hou, Sensitivity of measuring the urban form and greenery using street-level imagery: A comparative study of approaches and visual perspectives, Int. J. Appl. Earth Obs. Geoinf. 122 (2023) 103385.

SCAN ME

Personal Homepage

SCAN ME

Full Paper